

Physics XI - Hydraulics

Time Allowed : 1 hour _____ **Maximum Marks : 60**

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. This question paper consists of 4 questions carrying 10 marks each , while Q5 carries 20 marks.

Q1: Explain why

- (a) The blood pressure in humans is greater at the feet than at the brain
- (b) Atmospheric pressure at a height of about 6 km decreases to nearly half of its value at the sea level, though the height of the atmosphere is more than 100 km
- (c) Hydrostatic pressure is a scalar quantity even though pressure is force divided by area.

Q2: A U-tube contains water and methylated spirit separated by mercury. The mercury columns in the two arms are in level with 10.0 cm of water in one arm and 12.5 cm of spirit in the other. What is the specific gravity of spirit ?

Q3: In the previous problem, if 15.0 cm of water and spirit each are further poured into the respective arms of the tube, what is the difference in the levels of mercury in the two arms ? (Specific gravity of mercury = 13.6)

Q4: Two vessels have the same base area but different shapes. The first vessel takes twice the volume of water that the second vessel requires to fill upto a particular common height. Is the force exerted by the water on the base of the vessel the same in the two cases ? If so, why do the vessels filled with water to that same height give different readings on a weighing scale ?

Q5: A Pitot tube (Fig.) is mounted along the axis of a gas pipeline whose cross-sectional area is equal to S . Assuming the viscosity to be negligible, find the volume of gas flowing across the section of the pipe per unit time, if the difference in the liquid columns is equal to Δh , and the densities of the liquid and the gas are ρ_o and ρ respectively.

